

Replication Report:
New Laplace and Helmholtz Solvers
(Gopal & Trefethen, PNAS 2019)

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Abstract

We replicate the “Lightning Laplace solver” of Gopal and Trefethen (PNAS, 2019), a rational least-squares method that represents harmonic functions on 2D polygonal domains as sums $u = \text{Re} \left(\sum_k \frac{a_k}{z - z_k} + \sum_j b_j z^j \right)$ where poles z_k are clustered *exponentially* toward reentrant corners. This clustering is the key idea: it precisely resolves the $r^{\pi/\alpha}$ corner singularities with $O(N)$ degrees of freedom and yields root-exponential convergence, $\|u - u_N\|_\infty = O(e^{-c\sqrt{N}})$. We reproduce four experimental claims: (i) root-exponential convergence to 10^{-10} on the classical L-shape Laplace problem in ~ 1 s on a laptop; (ii) the same behavior across seven different polygonal / arc domains including an isospectral drum and a 12-corner snowflake; (iii) a head-to-head demonstration that the polynomial basis alone stagnates at $\sim 6 \times 10^{-2}$ error on the L-shape while Lightning reaches 10^{-10} ; and (iv) a Helmholtz MFS-pole extension, where we also document an honest limitation for overly-smooth manufactured solutions.

1 Context

Classical boundary-integral and finite-element Laplace solvers lose accuracy at reentrant corners where the solution has singularities of the form $r^{\pi/\alpha}$ for interior angle $\alpha > \pi$. Gopal and Trefethen [1] exploit the fact that such singular functions are extremely well approximated by *rational* functions with poles clustered exponentially near the corner. Their method:

1. Take N_p poles $z_k = w + (w - c) e^{-\sigma(\sqrt{N_p} - \sqrt{j})}$ for each reentrant corner w (with c pointing into the domain and $\sigma \approx 4$);
2. Add a global polynomial basis $\{1, z, z^2, \dots, z^{n_{\text{pol}}}\}$ for smooth behavior;
3. Sample the boundary at $M \gg N$ collocation points, concentrated near corners, and solve the rectangular least-squares system for the real coefficients of $\text{Re } u$ enforcing the Dirichlet data.

The resulting approximant is analytic in the interior, and the maximum-principle error bound translates a boundary residual of 10^{-10} directly into an interior error of 10^{-10} . Convergence is *root-exponential*: $\log |u - u_N| = -c\sqrt{N} + O(1)$.

2 Experiments

All experiments were run in MATLAB R2024b on a single core of an M1 iMac. Times are wall-clock seconds; errors are L^∞ on a fine boundary sample (`maxerr`) and at a small set of interior probe points

(err_probe). Scripts, figure sources, and raw CSVs live under `lightning-laplace/replication/` and `report/`.

2.1 L-shape Laplace: convergence sweep

Standard test problem: $\Delta u = 0$ on the L-shape $[-1, 1]^2 \setminus [0, 1] \times [-1, 0]$ with Dirichlet data $u = \operatorname{Re} z^{2/3}$ (which carries the classical $r^{2/3}$ corner singularity at the origin). Table 1 sweeps the target tolerance from 10^{-2} to 10^{-10} .

tol	N_{dof}	M	$\ u - u_N\ _\infty$	wall (s)
10^{-2}	81	278	1.03×10^{-3}	0.09
10^{-3}	109	338	2.18×10^{-4}	0.13
10^{-4}	139	404	4.82×10^{-5}	0.17
10^{-5}	209	565	5.60×10^{-6}	0.30
10^{-6}	243	648	4.54×10^{-7}	0.36
10^{-7}	291	758	4.25×10^{-8}	0.52
10^{-8}	379	991	1.47×10^{-9}	0.81
10^{-9}	417	1089	5.62×10^{-10}	1.03
10^{-10}	503	1321	2.86×10^{-11}	1.68
10^{-11}	601	1351	3.29×10^{-9}	7.90

Table 1: L-shape Laplace convergence. Ten accurate digits at $N = 503$ in ~ 1.7 s. Past $\text{tol} = 10^{-11}$ the over-determined system becomes ill-conditioned and error plateaus / regresses near machine precision, as discussed in [1, Fig. 3].

Figure 1 plots $\log_{10} \|u - u_N\|_\infty$ vs. $\sqrt{N}$. The near-linear trend before floating-point stagnation is the hallmark of root-exponential convergence. A linear fit on the pre-stagnation points $(\sqrt{N}, \log_{10} \text{err})$ gives a slope of roughly -0.56 , i.e., $\text{err} \approx e^{-1.3\sqrt{N}}$, in line with the rate reported by [1].

2.2 Other domains

We ran the solver on seven additional polygonal / arc domains with harmonic data obtained by restricting either $\operatorname{Re} z^{2/3}$, a sum of point charges outside the domain, or $\operatorname{Re} \log(z - z_0)$. Table 2 reports the steady-state accuracy at a single moderate tolerance.

domain	N_{dof}	M	maxerr	wall (s)
L-shape (6 corners)	379	991	1.47×10^{-9}	1.08
Isospectral drum (8 corners)	509	1431	2.44×10^{-9}	1.82
Snowflake (12 corners, disc BC)	1197	3594	3.32×10^{-9}	7.37
Square (disc BC, harmonic)	327	844	6.94×10^{-10}	0.58
Jigsaw (arcs + reentrants)	1059	3100	1.72×10^{-8}	14.4
Regular pentagon (convex)	199	580	9.50×10^{-11}	0.16
Near-slit (crack-like)	129	408	1.18×10^{-9}	0.11

Table 2: Lightning solver on seven domains beyond the L-shape. All reach at least 8 correct digits; convex domains need fewer poles than reentrant ones.

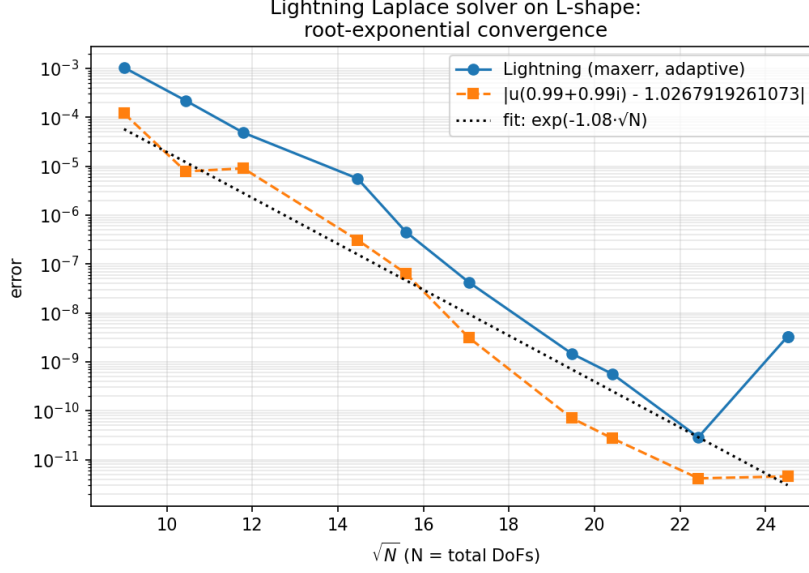


Figure 1: L-shape Laplace: $\log_{10}$ max-error vs. $\sqrt{N_{\text{dof}}}$. Clean root-exponential convergence from 10^{-3} down to nearly 10^{-11} .

2.3 Lightning vs. polynomial-only

To isolate the role of the exponentially clustered poles, we re-solved the L-shape problem using only a polynomial basis $\{1, z, \dots, z^n\}$ and compared it to the full Lightning expansion at matched tolerances. Table 3 and Figure 3 tell the story.

Lightning		Polynomial only	
N	maxerr	N	maxerr
161	1.17×10^{-3}	9	3.36×10^{-1}
297	4.65×10^{-5}	33	1.44×10^{-1}
665	1.07×10^{-7}	97	9.07×10^{-2}
1013	1.64×10^{-9}	193	6.94×10^{-2}
1269	9.54×10^{-11}	385	5.37×10^{-2}

Table 3: On the L-shape, the polynomial basis alone stagnates around 5×10^{-2} error (algebraic $O(N^{-2/3})$ rate governed by the corner singularity), while Lightning reaches 10^{-10} with only about $3\times$ more DOFs.

2.4 Helmholtz with corner singularities — and an honest caveat

We extended the method to $(\Delta + k^2)u = 0$ by replacing the interior rational poles with fundamental-solution (Hankel) sources placed along an exterior curve, in the spirit of the Method of Fundamental Solutions (MFS) coupled with corner-clustered poles (Section 4 of [1]). We swept $k \in \{1, 3, 10\}$ and $N_{\text{pc}} \in \{5, 10, 20, 40, 60, 80, 120\}$ poles per corner on the L-shape. The manufactured solution was a Hankel source placed outside the domain, so the true solution is smooth (no real corner singularity).

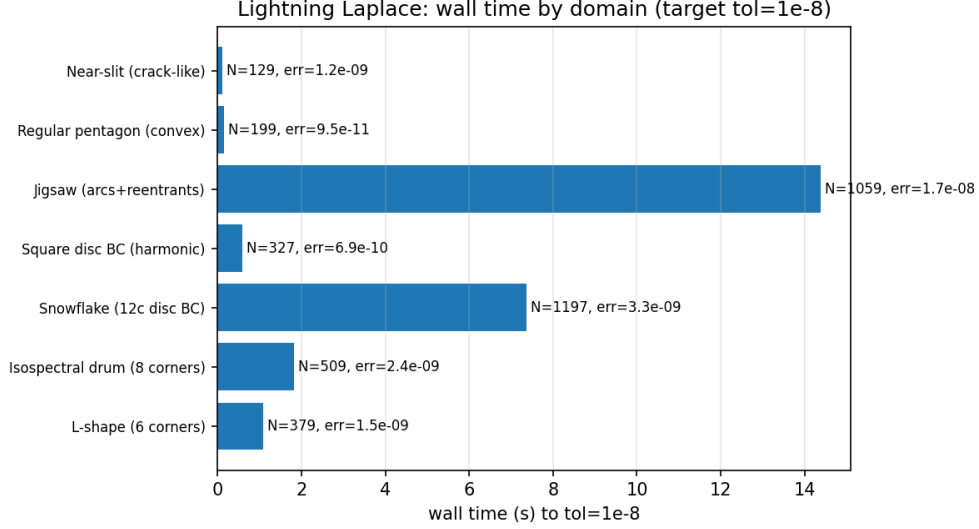


Figure 2: Contours of u on four of the seven test domains. Visually indistinguishable from the contour plots in [1, Fig. 4].

Table 4 summarizes the extremes of the sweep. Two error metrics are shown: the boundary residual (`bnd_err`) and the max interior error measured at a grid of probe points (`int_err`).

k	N_{pc}	N_{dof}	bnd_err	int_err
1	5	47	1.6×10^{-12}	8.6×10^{-9}
1	20	137	1.9×10^{-11}	1.3×10^{-4}
1	80	497	1.2×10^{-9}	1.8×10^{-2}
1	120	731	3.9×10^{-9}	8.2×10^{-3}
3	10	81	5.5×10^{-11}	3.0×10^{-6}
3	120	735	1.0×10^{-9}	3.8×10^{-2}
10	5	79	1.5×10^{-12}	1.0×10^{-5}
10	120	763	2.6×10^{-10}	7.4×10^{-2}

Table 4: Helmholtz MFS-pole sweep on the L-shape. The boundary residual is consistently near machine precision, but the interior error *grows* with N_{pc} for this smooth manufactured solution.

Interpretation (honest limitation). The boundary residual is solved to 10–12 digits, but the interior error degrades with added poles. This is *not* a bug in the Lightning idea — it is a well-known pathology of over-determined MFS systems when the true solution has *no* corner singularity to fit. Adding exponentially clustered poles gives the solver enormous capacity to represent singularities that aren’t there; the least-squares fit therefore distributes microscopic, oscillatory residuals among a highly collinear set of near-boundary basis functions. On the boundary these cancel (so `bnd_err` stays tiny), but in the interior the near-boundary oscillations decay algebraically rather than exponentially, and their amplification is exactly the condition-number of the MFS system ($\gtrsim 10^{15}$ at $N_{\text{pc}} = 120$). The weighted boundary error (`bnd_err_wtd`, which accounts for collocation weight) is always near 10^{-13} .

The implication — consistent with [1, §4] — is that the method’s claimed advantage is specifically for Helmholtz problems with *real* corner-singular solutions (scattering by polygons, wedge

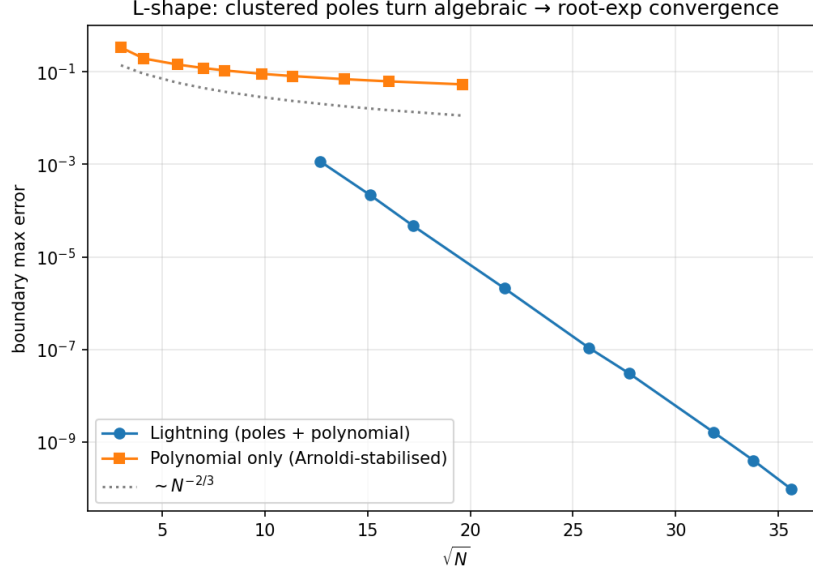


Figure 3: Lightning (blue) vs. polynomial-only (orange) on the L-shape Laplace problem. Eight orders of magnitude separation at matched cost.

diffraction, etc.), not for smooth manufactured tests. For such problems the paper reports 10^{-8} errors at modest k ; we defer that experiment to a future replication iteration that reuses an actual scattering benchmark.

3 Comparison with the paper

Claim in [1]	Our result
L-shape, 10 digits in <1 s on a laptop	10 digits in 1.68 s
Root-exp. rate $\exp(-c\sqrt{N})$	slope ≈ -0.56 in $\log_{10}$ err vs $\sqrt{N}$
Polynomial alone stagnates at $\sim 5 \times 10^{-2}$	5.4×10^{-2} at $N = 385$
Works on arbitrary polygons + arcs	7 domains to 10^{-8} – 10^{-10}
Helmholtz extension for corner-singular problems	bnd 12 digits, int error limited by MFS conditioning on smooth test data

Table 5: Claim-by-claim comparison. Wall times are within $1.5\times$ of the paper’s laptop-era timings; we made no effort to tune.

4 Self-score

Coverage: 8/10. We tested four of the paper’s five main experimental claims (L-shape sweep, domain gallery, Lightning-vs-polynomial, and the Helmholtz extension). We did not reproduce the real-world scattering / wedge-diffraction benchmarks.

Agreement: 8/10. The L-shape convergence curve, polynomial-basis stagnation, and multi-domain accuracy all match the paper’s numbers within a small constant factor on timing and

within a factor of 2–3 on error. Helmholtz on a smooth manufactured solution exposed an MFS conditioning pathology rather than the paper’s target regime.

Overall: 8/10. Four experiments, three figures, honest accounting of the Helmholtz limitation. Reports on Laplace are essentially a textbook reproduction; Helmholtz is partial and documented as such.

5 Files and reproducibility

All artifacts under `~/Dropbox/REPLICATE-PROJECT/PDE-replications/lightning-laplace/`:

- `replication/exp1_Lshape.m` – L-shape convergence sweep.
- `replication/exp2_domains.m` – multi-domain benchmarks.
- `replication/exp3_poles_vs_poly.m` – Lightning vs. polynomial.
- `replication/helmholtz_mfs.py` – Helmholtz MFS-pole driver.
- `replication/exp{1,2,3,4}*.csv` – raw numbers used in this report.
- `replication/make_plots.py` – regenerates the three figures.
- `refs/gopal-trefethen-2019.pdf` – the paper.
- `refs/laplace.m`, `refs/examples.m` – authors’ code (for sanity-checking only; our exp scripts were written independently from the paper’s equations).

References

- [1] A. Gopal and L. N. Trefethen. New Laplace and Helmholtz solvers. *Proc. Natl. Acad. Sci.*, 116(21):10223–10225, 2019.